1. THE AI LANDSCAPE

- Trace the six-stage evolution of AI from rule-based AI to human-AI collaboration.
- What distinguishes rule-based AI from machine learning?
- Explain how deep learning differs from conventional machine learning.
- Define generative AI and give examples of the kinds of outputs it can produce.
- What makes agentic AI different from a one-prompt-at-a-time chatbot?
- Describe the plan-act-observe-revise idea underlying agentic AI.
- A research agent automatically deletes a data file after incorrectly deciding it is duplicated. Why is this more serious than a wrong chatbot answer, and what control should be added?
- Name the five specialist roles shown in the multi-agent system and state the function of each.
- Why can specialisation improve a multi-agent system while coordination still determines reliability?
- A Researcher agent invents a citation, the Coder uses it to select a dataset, and the Reviewer checks only the code. Analyse the failure.
- A researcher wants one model to read a paper's text, inspect a medical image, and compare the image with the written report. What kind of model is needed, and why?
- Compare a text-only model with a multimodal model for interpreting a research paper containing figures and tables.
- Explain how reasoning-oriented models differ from traditional one-pass models.
- Why does the use of a reasoning-oriented model not eliminate hallucination?
- Differentiate quantisation, distillation, pruning, and LoRA as techniques for efficient AI.
- Explain the main trade-offs between large cloud models and small edge models shown.
- A lab must analyse sensitive interview recordings where internet connectivity is unreliable. Which deployment direction is preferable and why?
- A university is planning AI projects in drug discovery, battery materials, medical imaging, and crop forecasting. How does this portfolio illustrate AI for science and engineering?
- Describe the four-stage sense-understand-plan-act-and-learn cycle of physical and embodied AI.
- Compare the risk profile of AI in software engineering with AI in cybersecurity as presented.
- What are the six dimensions of trustworthy AI?
- Define sovereign AI.
- Explain the meaning of the expression 'AI Capability × Human Judgement = Responsible Impact'.

2. FOUNDATIONS OF AI-ASSISTED RESEARCH

- What three levels explain why AI is used in research?
- Explain why the research bottleneck is shifting from finding information to evaluating information.
- Compare traditional, digital, data-driven, AI-assisted, and agentic research practice.
- Why is the research lifecycle described as iterative rather than linear?
- Name the four broad stages in the research lifecycle shown.
- A student asks an AI system to choose the final research problem entirely on its own. What responsibility has been improperly transferred?
- Explain the division of labour between AI's potential strengths and human responsibilities in research.
- Compare AI as a tool, AI as an assistant, and AI as a collaborative system.
- Why should verification become more systematic as AI autonomy increases?
- What does it mean to treat AI output as an 'intermediate research object'?
- Differentiate AI-assisted research from AI-generated research.
- A researcher asks an AI system to write the complete discussion section from model outputs and submits it without checking causal claims. Identify the problem and a better practice.
- Define vibe coding.
- Why is vibe coding especially risky for research software if code is accepted without inspection?

- A generated analysis script executes without an error but silently uses the wrong column for the target variable. Which research principle is violated and what should the student do?
- Compare human epistemic responsibility with computational assistance in research.
- A Ph.D. scholar uses AI to summarise 100 papers, but personally checks only the five papers that support the preferred hypothesis. What is wrong with this workflow?
- Explain why the statement 'the ability to assist does not imply the authority to decide' is central to AI-assisted research.

3. AI LITERACY, LLMS, GROUNDING AND HALLUCINATION

- What are the four components of AI literacy presented?
- Why is simply using an AI tool not the same as being AI literate?
- Describe the simplified LLM workflow from prompt to token prediction.
- What is a token, and why do tokens matter when using LLMs for research?
- Compare an LLM's internal learned knowledge with a database lookup.
- A student pastes a very long thesis into an LLM and later notices that early chapters are ignored in the answer. Which concept explains this and what should the student do?
- Explain the problem created by a knowledge cut-off when asking about a newly published paper.
- Define grounding in AI-assisted research.
- Explain the four-stage RAG process.
- Compare RAG with GraphRAG.
- Why does RAG improve verifiability without guaranteeing correctness?
- A RAG system retrieves a relevant article but the generated answer reverses the paper's conclusion. Identify the failure and the required verification step.
- What three forms of hallucination are highlighted?
- A chatbot provides a realistic-looking DOI for a paper that cannot be found. What should the student do?
- Differentiate a wrong fact, an invented citation, and a wrong interpretation.
- Outline a four-step procedure for verifying a citation.
- Why are automation bias and overconfidence especially dangerous in scholarly work?

4. PROMPT ENGINEERING FOR RESEARCH

- What is the R-C-S-E-O-V framework for research prompting?
- Explain why context is described as one of the most powerful parts of a research prompt.
- Explain why verification should be designed into a prompt rather than added only after generation.
- Compare a weak research prompt such as 'Find papers on concept drift' with a strong research prompt.
- A student asks an AI assistant, 'Give me 20 references on AI in medicine,' and plans to paste them into a paper. Identify the prompt-design problem and the research risk.
- Rewrite the weak prompt 'Find papers on medical AI' using the R-C-S-E-O-V framework.
- Improve the prompt 'Tell me the research gap in image dehazing' so that the AI does not declare a gap without evidence.
- Rewrite 'Summarise these papers' into a research-grade prompt that produces a comparable literature matrix.
- Improve the prompt 'Which paper is best?' so that it supports scholarly comparison rather than an unsupported ranking.
- Rewrite 'Give me recent papers on concept drift' to include scope, evidence, and verification requirements.
- A researcher wants an AI system to extract claims from ten papers. Write a prompt that forces a claim-to-source evidence chain.
- Rewrite 'Explain this dataset' so that the output is suitable for planning a research analysis.
- Improve the prompt 'Make a graph from my data' using R-C-S-E-O-V principles.
- Rewrite 'Write my discussion section' into a responsible AI-assisted writing prompt.

- A student has a list of AI-generated references. Write a prompt for a verification assistant that implements the evidence chain.

5. EVALUATING AI RESEARCH TOOLS

- What three questions should a researcher ask when evaluating any AI research tool?
- What is meant by the 'source base' of an AI research tool?
- Explain why a polished interface is insufficient for judging a research tool.
- Compare LLM internal knowledge, web search, scholarly databases, citation indexes, and knowledge graphs as research sources.
- Why are scholarly databases and knowledge graphs preferred over unconstrained LLM knowledge for many research tasks?
- A chatbot gives a concise answer with no sources, while a scholarly search engine returns papers with traceable metadata. Which is more appropriate for building a literature review evidence base, and why?
- List the six dimensions in the Tool Evaluation Scorecard.
- Explain why freshness is a separate evaluation dimension from coverage.
- Compare citation traceability with transparency in the tool scorecard.
- A new literature tool has excellent coverage but does not reveal its source corpus or provide links to original papers. Identify the weaknesses using the scorecard.
- A researcher needs publication trends, institutional comparisons, and collaboration networks from structured scholarly data. Which source type is most suitable and why?
- A researcher needs grey literature and current webpages about a rapidly changing technology policy issue. Which source type is useful, and what limitation must be controlled?
- A team needs to analyse who cites whom and how an idea spreads through a field. Which source type should be prioritised?
- A student chooses a tool because it is popular on social media despite weak citation traceability. What principle of research-tool selection is being ignored?
- What does reproducibility mean when evaluating an AI research tool?

6. LITERATURE DISCOVERY AND REVIEW

- Differentiate literature search from literature discovery.
- Why can keyword search miss important literature even when the researcher chooses reasonable terms?
- What is a seed paper?
- Explain backward and forward citation searching.
- Compare co-citation and bibliographic coupling.
- Why does citation count indicate influence rather than necessarily quality or relevance?
- A student excludes a strong 2025 paper because it has only 25 citations and retains a 2015 paper with 5,000 citations solely because of the larger count. Analyse the mistake.
- A review team wants AI to speed up its workflow but not replace scholarly judgement. Which literature-review tasks can reasonably be accelerated?
- What parts of a literature review should AI not replace?
- Why is a single prompt such as 'generate my literature review' an inadequate scholarly workflow?
- Describe the integrated discovery workflow 'Define → Discover → Expand → Map → Analyse'.
- A researcher wants to rapidly understand the structural neighbourhood around one important paper using a similarity graph. Which tool is appropriate and why?
- A Ph.D. scholar wants to keep watching a topic over several years and be alerted to newly relevant papers. Which tool is appropriate?
- A researcher wants an AI-powered scholarly search entry point and then plans to explore references, citations, and authors. Which tool is suitable?
- A team wants to analyse publication trends and collaboration networks programmatically using an open scholarly knowledge graph. Which tool should it consider?

- A researcher has a small trusted collection and wants to visually expand the citation network around it over time. Which tool is suited to this task?
- Compare Semantic Scholar, Research Rabbit, Connected Papers, OpenAlex, and Litmaps by their primary role in a literature workflow.
- Why does tool orchestration matter more than simply collecting many AI tools?
- A scholar relies only on ChatGPT to discover literature for a systematic review. Identify the methodological weakness and propose a better approach.
- What is meant by the rule “search before synthesis” in AI-assisted literature review?
- Why should important citations be verified in the original paper rather than accepted from an AI summary?
- A generated literature review states, 'No prior work has addressed this problem,' but provides no systematic evidence. How should a researcher evaluate this claim?
- Compare prior works, derivative works, and similar papers when expanding from a seed paper.
- What is the value of using multiple discovery strategies instead of one tool or one query?
- Design a defensible literature-discovery workflow for a new Ph.D. topic using concepts.

7. DATA VISUALIZATION PRINCIPLES

- What are the four levels in Munzner's Four-Level Nested Model?
- A team builds a technically elegant dashboard before speaking to intended users and later discovers it answers the wrong questions. Which level failed first?
- Differentiate categorical, ordinal, and quantitative attributes.
- What are sequential, diverging, and cyclic ordering directions?
- Define lookup, browse, locate, and explore as search tasks in visualization.
- Why is writing a task as an {action, target} pair useful?
- Rank the visual magnitude channels from most to least accurately perceived.
- Explain expressiveness in visual encoding.
- Explain effectiveness in visual encoding.
- Compare expressiveness and effectiveness.
- A student uses a pie chart to compare five nearly equal model accuracies. What would you recommend and why?
- Differentiate sequential, diverging, categorical, and cyclic colour palettes and give a suitable use for each.
- Why should a visualization not rely only on red-versus-green distinctions?
- What four grouping methods can be used in visualization design?
- Compare grouping by proximity with grouping by similarity.
- Explain the four design principles: proximity, alignment, repetition, and contrast.
- A chart encodes a quantitative variable only by different hues with no ordered luminance. Which principle is violated and how should it be corrected?
- Compare the domain-situation level with the algorithm level in Munzner's model.

8. AUTOMATED VISUALIZATION AND LIDA

- What limitations of earlier automated-visualization approaches motivated systems such as LIDA?
- Compare heuristic, task-decomposition, and learning-based approaches to automated visualization.
- What is LIDA,?
- Why is LIDA described as grammar-agnostic?
- Name the four main LIDA modules and state their roles.
- Explain the two-stage role of the Summarizer in LIDA.
- What does the Goal Explorer produce?
- Compare the roles of Goal Explorer and VisGenerator.
- What is fill-in-the-middle generation, and why can a scaffold reduce structural errors?
- Explain compile checking, self-consistency, and calibration as candidate-filtering mechanisms.

- How does temperature affect candidate generation in LIDA-like workflows?
- A generated chart compiles successfully but uses the wrong variable on the y-axis. Why is compile checking insufficient?
- Why can LIDA's Infographer create an additional research risk?
- Define Visualization Error Rate (VER).
- A system generates 285 charts and 16 contain code errors. Calculate VER.
- A system generates 500 charts and 25 fail because of code errors. Calculate VER.
- A visualization system has a VER of 4% over 750 generated charts. How many charts contained code errors?
- An automated visualization system asks an LLM judge to score each generated chart for code accuracy, data transformation, goal compliance, visualization type, data encoding, and aesthetics. Which evaluation approach is being used, and what limitation should the researcher remember?
- Why is data summarisation essential grounding for LIDA?

9. AI RESEARCH TOOL ECOSYSTEM

- Summarise the major eras in the evolution of AI for science.
- Compare foundation models, instruction-tuned LLMs, and agentic/multimodal AI in the AI-for-science timeline.
- A research group uses one general chatbot for literature search, academic writing, qualitative coding, domain modelling, and workflow monitoring. How should the group rethink its tool strategy using an AI research tool ecosystem view?
- Explain why a researcher should think in terms of tool categories rather than treating every AI tool as interchangeable.
- A researcher needs broad help with reasoning, writing, and coding but will separately verify scholarly evidence. Which category of tools is appropriate?
- A student wants help improving academic language and style rather than discovering literature. Which tool category should be selected?
- A team is analysing interview transcripts qualitatively. Which category and examples are relevant?
- A materials-science team wants a domain-specific AI system rather than a general chatbot. Which example fits this category?
- Compare general-purpose assistants with domain-specific research tools.
- What is the purpose of LLMOps and infrastructure tools in the research ecosystem?
- Explain the purpose of smart citations as described for scite.ai.
- What does FAIR stand for in FAIR data?
- Compare Findable and Accessible with Interoperable and Reusable in FAIR data.
- Why can local AI be important for research involving patient data or interview transcripts?
- What is the difference between MCP and A2A?
- A lab wants an agent to access a database and a plotting tool. Another lab wants two specialised agents to delegate tasks to one another. Which integration idea applies to each?
- What role does the Eleven-Layer Unified AI Research Framework play in the AI research tool ecosystem?
- Why should an AI research ecosystem include both end-user tools and infrastructure/evaluation tools?
- Explain why local model runners such as Ollama, LM Studio, LocalAI, and llama.cpp are conceptually different from cloud-based general assistants.

10. RISKS, ETHICS AND GOVERNANCE

- What eight major challenges arise in AI-assisted research?
- Define algorithmic bias in the context of research discovery systems.
- How does bias and fairness in training data differ from algorithmic bias in recommendation systems?
- Why is lack of explainability particularly important in high-stakes research applications?
- What is 'shadow AI' as a data-privacy risk in research?

- A researcher uploads identifiable patient data to a public AI service without checking ethics approval or data-processing rules. Identify the risks and the appropriate response.
- Explain why reproducibility is difficult with generative AI and how researchers can improve it.
- What is meant by over-reliance or cognitive atrophy in AI-assisted research?
- Why can stylistic homogenisation be a scholarly concern?
- Describe the Ethical AI Loop.
- What information should an AI-use disclosure statement include?
- Explain why disclosure does not transfer responsibility from the authors to the AI tool.
- Why should an AI system not be listed as an author of a research paper?
- A student uses AI to improve grammar in two sections and generate plotting code for one figure. What should a suitable disclosure communicate?
- What four major governance concerns should be considered in AI-assisted research?
- A laboratory replaces most junior researchers' formative coding and literature-screening tasks with automation. What governance concern arises, and how should it be managed?
- Why are environmental costs relevant to AI governance in research?
- What is dual use, and why do autonomous agents require access controls?
- A research agent can send emails, modify cloud files, and run code. Propose a minimal governance design for this system.
- A researcher must choose between a convenient cloud AI service and an approved local model for confidential interview transcripts. Which option better protects research privacy, and why?